BAM285 – BUSINESS ANALYSIS IN IT

FINAL COURSEWORK

EMPLOYEE LEAVE AND VACATION REQUESTS AUTOMATION SYSTEM

A Business Process Automation Plan for Employee Leave and Vacation Requests

Prepared by:

Marcel Jablo

Wayne Blando

Aaron Lagaday

Joemarie Balinton

Joseph Lim

BS Information Technology – 3rd Year

Section: BSIT 3-10

Course: BAM285 – Business Analysis in IT

Instructor: Jireh A. Flavio

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EXECUTIVE SUMMARY

The Employee Leave and Vacation Request Automation System is a proposed computerized system designed to make employee leave management faster, easier, and more organized. It allows employees to submit leave requests online while enabling supervisors and human resource personnel to review, approve, reject, and monitor applications efficiently. The system includes employee account management, request status tracking, and leave record management, with security measures to protect sensitive information.

The project aims to reduce paperwork, prevent errors, avoid delays, and improve transparency in managing employee absences. Through systematic development, testing, and implementation planning, the system is expected to improve administrative efficiency, support timely decision-making, and provide a more reliable and organized workplace.

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CHAPTER I – INTRODUCTION

1.1 Background of the Project

Employee leave and vacation automation is an important administrative function in every organization. Employees need time away from work for various reasons, including personal matters, illness, family responsibilities, emergencies, and scheduled vacations. To maintain an organized workplace, organizations must have a reliable process for submitting, reviewing, approving, and recording employee leave requests.

In organizations that rely on manual procedures, employees may need to complete paper forms and submit them to their supervisors or human resource personnel. This process can be time-consuming and may lead to misplaced documents, delayed approvals, communication problems, and inaccurate leave records. Monitoring employee absences and maintaining updated information can also become difficult, especially when the number of employees increases.

The Employee Leave and Vacation Request Management System is proposed to address these challenges through a computerized platform. The system will allow employees to submit leave applications electronically and monitor their request status. Authorized supervisors or administrators will be able to review applications, approve or reject requests, and maintain organized leave records.

Through this project, the organization can improve the efficiency of leave administration, reduce unnecessary paperwork, and provide a more convenient experience for employees and management.

1.2 Statement of the Problem

1.2.1 General Problem

The management of employee leave and vacation requests in an organization can be challenging due to manual processes, delays in approval, inaccurate record-keeping, scheduling conflicts, and a lack of proper monitoring. Supervisors may also forget to record or list approved leave requests, resulting in confusion about employee absences and possible disruptions to daily operations. These issues can affect workforce planning, communication between employees and supervisors, and the overall efficiency of leave management.

1.2.2 Specific Problems

1. Manual Leave Request Process – Employees may need to submit leave and vacation requests manually, making the process time-consuming and difficult to organize.

2. Delayed Approval of Requests – Supervisors may take a long time to review and approve or reject leave requests, causing uncertainty for employees.

3. Scheduling Conflicts – Multiple employees may request leave on the same dates, potentially resulting in insufficient staff to maintain daily operations.

4. Inaccurate Leave Balance Records – Employee leave credits and remaining vacation days may not be updated accurately, leading to discrepancies in leave records.

5. Lack of Request Status Updates – Employees may have difficulty knowing whether their leave requests are pending, approved, or rejected.

6. Difficulty Monitoring Employee Absences – Supervisors may struggle to monitor upcoming leaves and identify which employees will be absent on specific dates.

7. Miscommunication Between Employees and Supervisors – Leave requests and approval decisions may be overlooked, misunderstood, or not properly communicated.

8. Forgotten or Unrecorded Approved Leave Requests – Supervisors may forget to record or list approved leave requests in the schedule or monitoring records, causing approved absences to be overlooked and potentially leading to attendance discrepancies.

9. Difficulty Maintaining Organized Leave Records – HR personnel and supervisors may find it challenging to maintain, retrieve, and update employee leave records when information is scattered across paper forms, spreadsheets, or messages.

10. Difficulty Enforcing Leave Policies – Supervisors and HR personnel may struggle to ensure that leave requests comply with company policies, eligibility requirements, and available leave credits.

1.3 Objectives of the Project

1.3.1 General Objective

To develop a Web-Based Employee Leave and Vacation Request Management System that allows employees to submit leave requests online, enables management to approve or reject requests, and maintains organized records of all leave applications and their corresponding statuses.

1.3.2 Specific Objectives

The study specifically aims to:

1. To develop a secure login system that allows employees and management to access the website using their respective accounts.

2. To create an online leave request form that allows employees to provide the necessary details, such as leave type, dates, and reason for leave.

3. To develop an approval and rejection feature that enables management to review employee requests and approve or reject them.

4. To provide leave request status tracking so employees can check whether their requests are pending, approved, or rejected.

5. To automatically record submitted leave requests and management decisions in a centralized database for future reference.

6. To provide a leave request history feature that allows employees and authorized management personnel to view previous and current leave applications.

7. To prevent approved leave requests from being forgotten or omitted by automatically recording approved requests and displaying them in a management dashboard or leave schedule.

8. To improve the efficiency and accuracy of leave management by reducing manual paperwork, minimizing record-keeping errors, and organizing the entire request process through a single website.

1.4 Significance of the Project

The Employee Leave and Vacation Request Management System is expected to provide benefits to the following groups:

Employees. The system will provide employees with a convenient way to submit leave applications and monitor their request status without relying entirely on paper forms or repeated inquiries.

Supervisors and Managers. The system will help supervisors review applications, make approval decisions, and monitor employee absences more systematically.

Human Resource Personnel. The system will assist human resource personnel in maintaining organized leave records, retrieving application information, and preparing reports when necessary.

Organization. The proposed system can help reduce administrative workload, improve record accuracy, minimize paperwork, and support more efficient workforce planning.

Future Researchers and Developers. This project may serve as a reference for future studies and system development projects related to employee management, leave administration, and workplace automation.

1.5 Purpose and Description

The purpose of the Employee Leave and Vacation Request Automation System is to provide a centralized digital platform for managing employee leave and vacation applications. It is designed to make the process more convenient for employees and more manageable for supervisors and human resource personnel.

The system will allow employees to log in, complete leave request forms, specify the type and duration of leave, provide a reason when required, and submit applications for approval. Supervisors or authorized administrators will be able to view submitted requests, evaluate the information, and approve or reject applications based on organizational policies.

The system will also provide a way to monitor request status and maintain records of leave transactions. Depending on the final system requirements, additional functions such as automated leave balance tracking, and report generation may be included.

By centralizing leave information, the system aims to improve communication, reduce processing time, and support accurate recordkeeping within the organization.

1.6 Scope and Limitations

The project focuses on developing a system for employee leave and vacation request management. Its intended features include:

1. Employee account and profile management.
2. Electronic submission of leave and vacation requests.
3. Recording of leave types, dates, and reasons.
4. Review and approval or rejection of applications by authorized personnel.
5. Monitoring of application status.
6. Maintenance of employee leave request records.
7. Basic reporting of leave applications and transactions, if included in the final implementation.
8. Access control for employees and authorized administrators.

CHAPTER II – REVIEW OF RELATED LITERATURE AND SYSTEMS

2.1 Related Literature

Employee leave management is an important part of human resource administration because it helps organizations monitor employee absences, organize work schedules,

and maintain accurate personnel records. An effective leave management process allows employees to request time off while giving management sufficient information to make decisions based on staffing needs and organizational policies.

Traditional leave management often involves paper forms, manual recordkeeping, and direct communication between employees and supervisors. Although these methods may work in small organizations, they can become inefficient when the number of employees and leave applications increases. Manual procedures may also make it more difficult to retrieve historical records and identify scheduling conflicts.

Computerized information systems provide opportunities to improve administrative processes by storing information in an organized digital format. Through centralized records, authorized users can retrieve information more easily, reduce repetitive data entry, and monitor transactions more efficiently.

Web-based management systems can further improve accessibility by allowing users to interact with the system through compatible devices and web browsers. Depending on their design, these systems may provide user authentication, electronic forms, automated status updates, and report generation.

These concepts provide a foundation for developing the Employee Leave and Vacation Request Management System. By applying information technology to leave administration, the proposed project aims to reduce the difficulties associated with manual procedures and improve the management of employee leave applications.

2.2 Related System

Related systems include computerized employee management applications, online leave request platforms, and human resource information systems that provide functions for recording employee information and managing absences.

Manual Leave Request System. This approach uses paper forms that employees submit to supervisors or human resource personnel. It is relatively simple to implement but may require more time for document handling, record updating, and application tracking.

Electronic Leave Request System. This type of system allows employees to submit applications electronically. It can improve accessibility and reduce the need for printed forms. Depending on its features, it may also provide request status tracking and electronic approval.

Human Resource Information System (HRIS). An HRIS is a broader system used to manage employee-related information and administrative processes. Some HRIS platforms include leave management, employee records, attendance, payroll, and reporting functions. However, their available features and complexity vary by product.

The proposed Employee Leave and Vacation Request Management System will focus specifically on leave and vacation applications. Rather than attempting to replace every human resource function, it will provide the essential features required to submit, review, approve, and monitor leave requests in an organized manner.

2.3 Synthesis

The reviewed concepts and types of systems demonstrate the importance of accurate recordkeeping, accessible information, and organized approval procedures in employee leave management. Manual systems may be straightforward to use, but they can create administrative challenges when records must be updated frequently or multiple applications must be processed.

Electronic leave request systems offer a more organized alternative by providing digital submission, centralized records, and application status monitoring. Broader human resource information systems may offer additional features, but these may exceed the requirements of organizations that need a dedicated leave management solution.

Based on these observations, the proposed Employee Leave and Vacation Request Management System will focus on the essential functions of leave request submission, application review, approval processing, status tracking, and record management. These features are intended to address common difficulties in manual leave administration while providing a practical system that can be adapted to the organization's needs.

CHAPTER III – METHODOLOGY, RESULTS AND DISCUSSION

3.1 Requirements Analysis

3.1.1 Stakeholder Analysis

Stakeholder	Role	Needs
Employees	Submit leave and vacation requests	Easy request submission
Supervisors	Review and approve	workload balance
Human Resources (HR)	Validate entitlements	Accurate records
IT Department	Maintain system	System reliability

3.1.2 SIPOC Analysis

Suppliers	Inputs	Process	Outputs	Customers
Employees	Leave and vacation request details	Submit leave request	Submitted leave	Supervisors

			request	
Supervisors	Leave request and workload information	Review and approve or reject the request	Approval or rejection decision	Employees
Human Resources (HR)	Leave policies and employee leave records	Validate leave entitlements and update records	Accurate and updated leave records	Employees and Supervisors
IT Department	Leave management system and technical resources	Maintain the system and ensure it functions properly	Reliable and accessible leave managemen t system	Employees, Supervisors, and HR

3.1.3 Current-State Process

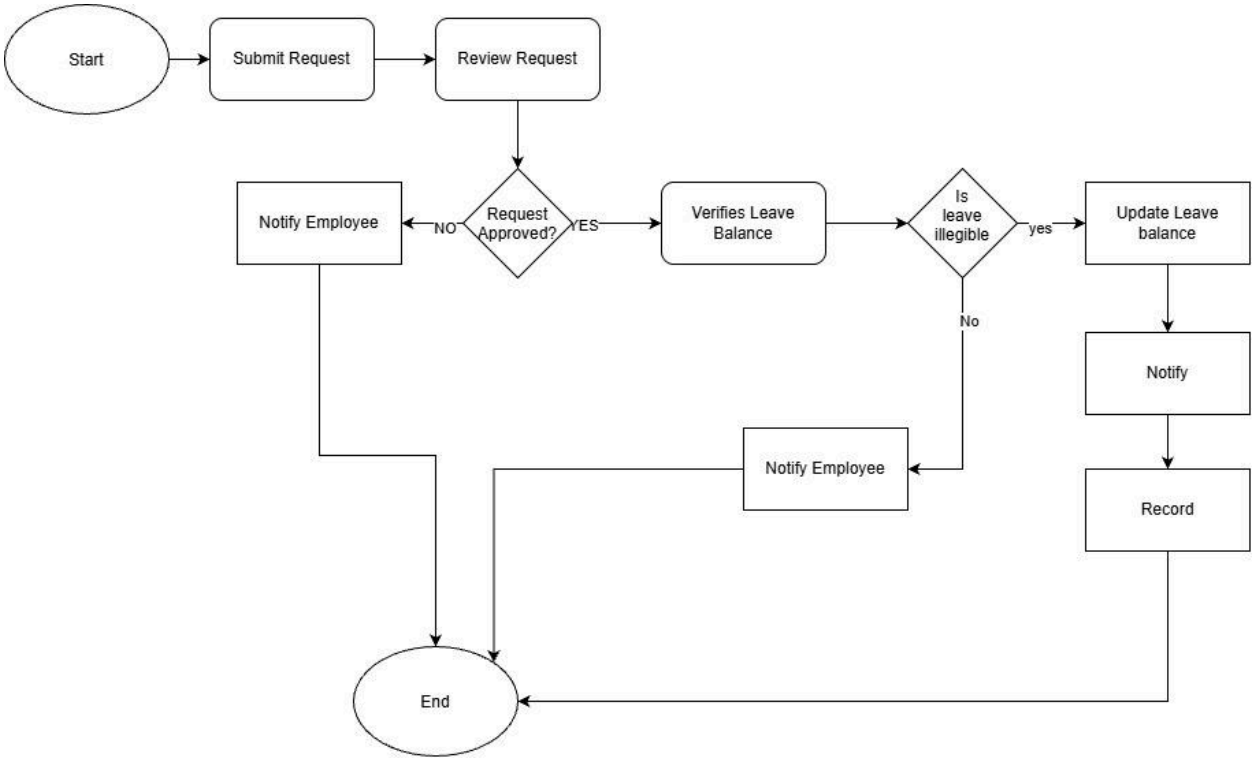


Figure 1. Current IT Support Process

3.1.4 Problems Identified

Problem	Root Cause	Effect
Delayed leave approval	Manual processing of requests	Employees wait longer for approval
Inaccurate leave balances	Lack of an automated tracking system	Incorrect calculation of remaining leave
Confusion about leave policies	Unclear rules and guidelines	Misunderstandings between employees and management
Lost leave requests	Poor recordkeeping and misplaced	Requests may be overlooked or

	forms	unprocessed
Scheduling conflicts	Lack of coordination between employees and managers	Staff shortages and disrupted operations
Delayed responses	Poor communication	Employee dissatisfaction
Unauthorized leave	Lack of a proper approval process	Attendance and staffing problems

3.1.5 Root-Cause Analysis

Problem: The root cause of employee leave and vacation request problems is the lack of a centralized management system, manual processing, unclear leave policies, and poor communication, resulting in delayed approvals, inaccurate leave records, scheduling conflicts, and employee dissatisfaction.

3.1.6 Requirements

Functional Requirements

ID	Requirement
FR01	The system shall allow employees to submit leave and vacation requests online.
FR02	The system shall allow managers to approve or reject leave requests.
FR03	The system shall allow employees to view the status of their leave requests.
FR04	The system shall automatically calculate and update employee leave balances.
FR05	The system shall notify employees about the approval or rejection of their requests.

FR06	The system shall allow HR personnel to manage employee leave records and schedules.
FR07	The system shall generate leave reports for management.

Non-Functional Requirements

Requirement	Description
Security	Security: The system shall protect employee information from unauthorized access.
Performance	Performance: The system shall process leave requests and display results quickly.
Usability	Usability: The system shall have a simple and user-friendly interface.
Reliability	Reliability: The system shall maintain accurate leave records and operate consistently.
Availability	Availability: The system shall be accessible to authorized users when needed.
Compatibility	Compatibility: The system shall work on common web browsers and devices.
Maintainability	Maintainability: The system shall be easy to update and maintain when requirements change.

3.2 Framework of the Project

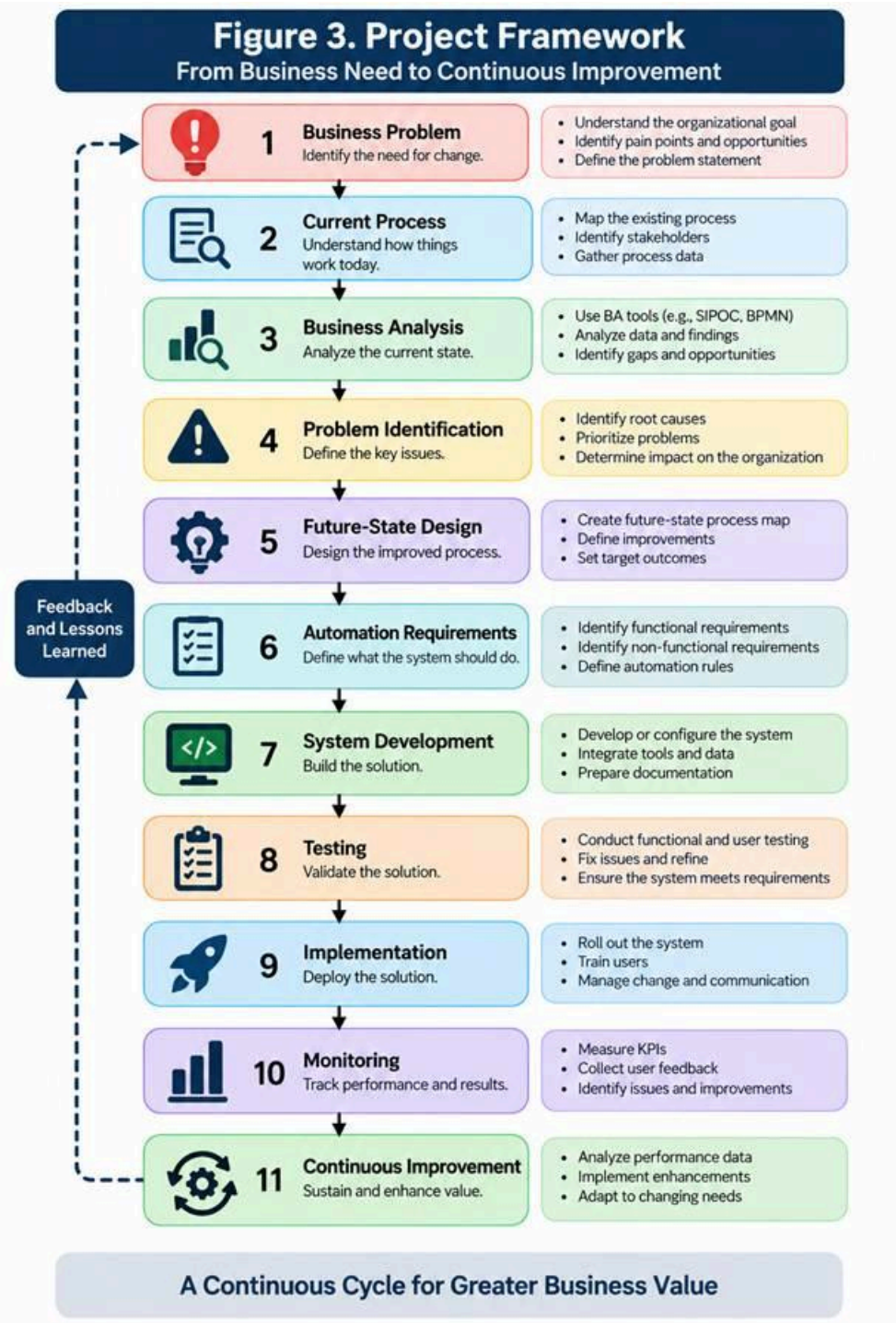


Figure 3. Project Framework

3.2.1 Proposed Future-State Process

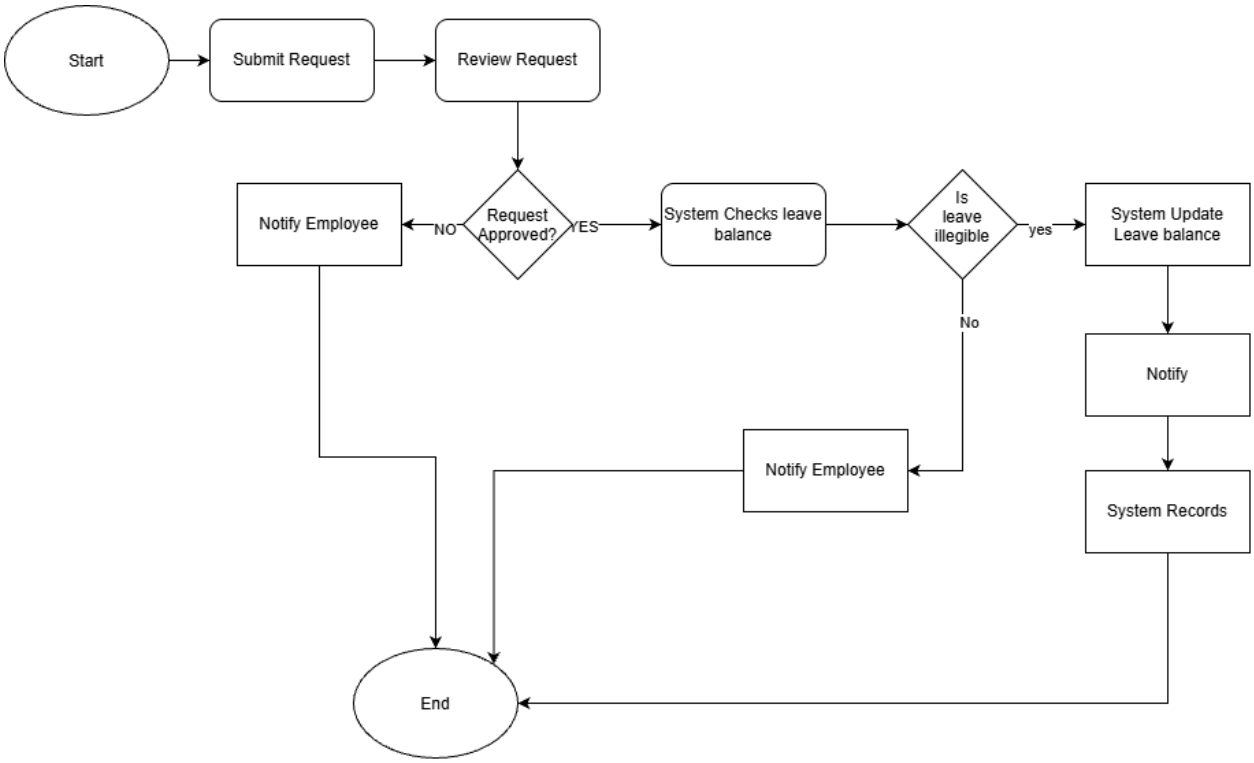


Figure 2. Proposed Automated Process

3.2.2 Before and After Comparison

Activity	Current Process	Proposed Process
Leave request submission	Manual	Online Form
Request approval	Manual	Automated Workflow
Request status tracking	Manual	Automatic
Leave balance monitoring	Manual	Automatic
Leave record management	Manual	Automatic
Schedule management	Manual	Automatic
Report generation	Inconsistent	Centralized

3.2.3 Proposed System Modules

Module 1 – User Management

Allows administrators to manage employees, IT technicians, IT managers, and user roles.

Module 2 – Requests Submission

Employees provide problem category, subject, description, location, urgency, and screenshot/attachment.

Module 3 – Requests Assignment

The system routes requests based on category, priority, and technician availability.

Module 4 – Requests Monitoring

Displays new, open, in-progress, pending, resolved, and closed requests.

Module 6 – Reporting

Reports include total requests, open/closed tickets, tickets by category/technician, and response/resolution time.

3.2.4 Automation Rules

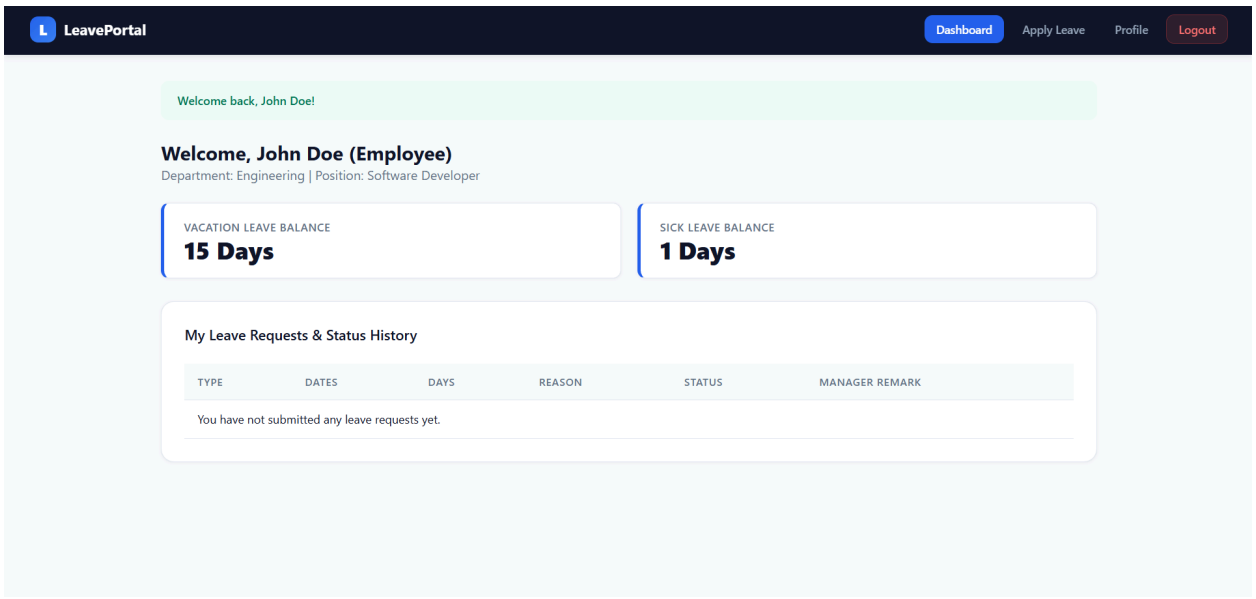
Rule	Condition	Automated Action
1	Leave request is submitted	Notify the manager or HR for review.
2	Leave balance is insufficient	Notify the employee that the leave credits are insufficient.
3	Leave request is approved	Notify the employee and update the request status to Approved.
4	Leave request is rejected	Notify the employee and provide the reason for rejection.
5	Multiple employees request leave on the same dates	Alert the manager about potential scheduling conflicts.

3.2.5 Sample Requests

Field	Sample Value
Request #	LR-2026-00125
Employee	Maria Santos
Department	Finance
Leave Type	Vacation Leave
Date Requested	October 23, 2026

Start Date	November 2, 2026
End Date	November 6, 2026
Number of Days	5 days
Subject	Vacation Leave Request
Description	I would like to request five days of vacation leave for a family trip.
Assigned Approver	Pedro Reyes
Status	Approved

3.2.6 Proposed Dashboard



3.2.7 Key Performance Indicators

KPI	Purpose
Number of Leave Requests	Measures total requests.
Average Approval Time	Measures approval speed.
Pending Requests	Tracks unapproved requests.
Rejected Requests	Counts rejected requests.
Leave by Type	Identifies common leave types.
Leave by Department	Tracks departmental requests.
Leave Balance Accuracy	Ensures accurate leave credits.
Approval Rate	Measures the percentage of approved requests.

3.3 Development and Testing

3.3.1 Prototype

Prototype Screen 1 – Login: IT Support System with username, password, and login button.

Prototype Screen 2 – Submit Requests: Category, priority, subject, description, attachment, and submit ticket button.

Prototype Screen 3 – My Requests: Ticket number, category, and status.

3.3.2 Test Cases

Test ID	Test	Expected Result	Result
TC01	Login using valid credentials	User successfully logs in	Passed

TC02	Submit leave request	Leave request is created	Passed
TC03	Submit incomplete request	System displays required fields	Passed
TC04	Check leave request number	Unique request number is generated	Passed
TC05	Approve leave request	Request status is updated to Approved	Passed
TC06	Reject leave request	Employee receives rejection notification	Passed
TC07	Update leave balance	Leave credits are updated correctly	Passed
TC08	Generate leave report	Leave report is successfully generated	Passed

3.3.3 Sample Test Result

Based on the sample testing, the proposed workflow successfully performed the primary ticket-management functions. The most important functions tested were ticket creation, ticket identification, ticket assignment, status update, resolution recording, ticket closure, and report generation.

The prototype should undergo additional user acceptance testing before actual deployment.

3.4 Implementation Plan

Phase	Major Activities	Output

Phase 1 – Preparation	Identify stakeholders, confirm requirements, document existing processes, and validate requirements	Approved requirements document
Phase 2 – System Design	Design workflow, database, user interface, and approval rules	System design and prototype
Phase 3 – Development	Develop leave request, approval, notification, and reporting modules	Functional prototype
Phase 4 – Testing	Conduct functional, integration, user acceptance, and security testing	Tested system
Phase 5 – Implementation	Deploy the system, register users, train employees and HR personnel, and monitor initial operations	Operational automated process
Phase 6 – Monitoring and Improvement	Monitor KPIs, collect feedback, analyze recurring problems, and improve workflows	Continuous improvement plan

Implementation Schedule

Activity	W1	W2	W3	W4	W5	W6
Requirements	✓					
Process Analysis	✓	✓				
System Design		✓	✓			
Development			✓	✓		
Testing				✓	✓	
Training					✓	
Deployment						✓
Monitoring						✓

Risk Analysis

Risk	Probability	Impact	Mitigation
Employees do not use the system	Medium	High	Conduct user training
Incomplete leave requests	Medium	Medium	Use required fields
System downtime	Low	High	Implement backup and recovery plans
Resistance to change	Medium	High	Provide orientation and communication
Incorrect leave balance	Medium	High	Validate leave calculations
Scheduling conflicts	Medium	Medium	Add schedule conflict

			alerts

Cost-Benefit Analysis

Item	Estimated Cost
Software Development	₱40,000
Hosting	₱15,000
Training	₱10,000
Maintenance	₱15,000
Contingency	₱10,000
TOTAL	₱90,000

These are illustrative figures for the sample project and are not actual organizational costs.

Expected Benefits

- Faster Processing
- Accurate Records
- Improved Communication
- Fewer Scheduling Conflicts
- Increased Efficiency
- Better Reporting
- Employee Satisfaction

3.4.1 Change Management Plan

The automation project for the employee leave and vacation request process should not focus only on technology. Employees, supervisors, and HR personnel must understand the purpose of the changes and how the automated system will improve the submission, approval, and monitoring of leave requests.

Change Strategy: Explain → Demonstrate → Train → Implement → Monitor → Improve

Employee Training

21. How to access the employee leave and vacation request system.
22. How to submit a leave or vacation request online.
23. How to select the appropriate leave category and enter the required details.
24. How to check available leave credits and monitor request status.
25. How to receive notifications regarding approval, rejection, or additional requirements.
26. How to cancel or modify a leave request, when permitted by company policy.

Supervisor and HR Personnel Training

27. How to receive and review employee leave requests.
28. How to verify leave eligibility, available leave credits, and supporting documents.
29. How to approve, reject, or return requests for correction.

30. How to monitor pending requests and manage approval deadlines.
31. How to update and verify employee leave records and balances.
32. How to generate reports on leave utilization, employee absences, and pending applications.

Implementation and Monitoring

Before full implementation, the organization should conduct a pilot test to identify potential issues and gather feedback from employees, supervisors, and HR personnel. After deployment, management should monitor system performance, processing time, approval delays, and user satisfaction. Feedback should be used to improve the automated process and ensure that the system meets the organization's operational requirements.

CHAPTER IV – RECOMMENDATIONS

1. Implement a Centralized Employee Leave and Vacation Request System All employee leave and vacation requests should be submitted through a centralized system to ensure that applications are properly recorded, tracked, and processed. This will help prevent lost requests and improve the efficiency of leave management.

2. Establish Standard Leave Categories and Policies Leave types, such as vacation leave, sick leave, emergency leave, and other applicable leave categories, should be clearly defined. Standardized policies will support consistent processing, eligibility verification, and accurate leave balance calculations.

3. Implement Automated Notifications and Approval Workflows The system should automatically notify employees, supervisors, and HR personnel whenever a leave request is submitted, approved, rejected, or requires further action. Automated approval routing and reminders can help reduce processing delays.

4. Establish Clear Approval Rules and Procedures The organization should establish clear guidelines for leave approval, including eligibility requirements, available leave credits, advance notice, staffing requirements, and escalation procedures for pending requests. These rules will promote fair and consistent decision-making.

5. Monitor Leave Management Performance Indicators (KPIs) Management and HR should regularly monitor key performance indicators, such as the number of leave requests submitted, average approval time, pending requests, leave balance accuracy, and employee absences. These indicators will help evaluate the effectiveness of the automated system.

6. Conduct Employee and HR Training Employees, supervisors, and HR personnel should receive proper training before the implementation of the automated leave and vacation request system. Training should cover request submission, approval procedures, leave balance monitoring, notifications, and system usage.

7. Continuously Improve the Leave Management Process The organization should regularly review leave request records, employee feedback, approval delays, and system performance to identify recurring issues and opportunities for further automation. Continuous improvement will help ensure that the system remains efficient, reliable, and responsive to organizational needs.

CONCLUSION

The analysis demonstrates that the existing IT support request process can be improved through business process automation.

The current process relies heavily on manual communication and tracking. This creates difficulties in maintaining centralized records, assigning requests, monitoring status, and generating reports.

The proposed IT Support Ticket Automation System introduces a standardized process in which requests are submitted, categorized, assigned, monitored, resolved, and reported through a centralized system.

The project demonstrates how business analysis can be used before implementing automation. Instead of simply developing software, the proposed approach begins with understanding the existing business process, identifying problems, defining requirements, designing the future-state process, and establishing performance measures.

The proposed automation plan therefore provides a structured approach for improving IT support request management.

APPENDIX A.1 – BIBLIOGRAPHY

For an actual student submission, this section should contain the real sources used during the research, formatted consistently according to the required citation style.

Example: Author/Organization. (Year). Title of article or publication. Publisher/Website.

The uploaded BAM285 module lists the UAGC Writing Center and PandaDoc business case study guidance among its references.

APPENDIX A.2 – RESOURCE PERSON

Field	Sample Information
Name	Trisha Pojas
Position	Employee
Organization	Accounting Company

Interview Questions

1. How do employees currently submit leave and vacation requests?
2. Who is responsible for reviewing and approving leave requests?
3. What problems do employees encounter when requesting leave?
4. How does HR track employee leave balances?
5. How are employees notified about approval or rejection?
6. How do managers handle overlapping leave schedules?
7. What features would you like to see in the proposed system?
8. How can the system improve the leave request process?

Sample Interview Finding

"The biggest problem is that employees submit leave and vacation requests through different channels, such as paper forms, emails, or direct messages. It is difficult to track pending requests, monitor leave balances, and ensure that all requests are reviewed and approved on time." "The biggest problem is that employees submit leave and vacation requests through different channels, such as paper forms, emails, or direct messages. It is difficult to track

pending requests, monitor leave balances, and ensure that all requests are reviewed and approved on time."

APPENDIX A.3 – PERSONAL VITAE

MARCEL JABLO

Field	Sample Information
Contact Information	mama.jablo.ui@phinmaed.com | 0994-047-4292
Educational Background	Bachelor of Science in Information Technology, PHINMA University of Iloilo, 2026–Present
Technical Skills	Python; Java; HTML/CSS; JavaScript; SQL; Database Management; Business Process Analysis; Process Mapping
Projects	Student Information System; IT Support Ticket Automation System; Inventory Management System